Cisco PE Node Optical Transport Failure Recovery Procedure for KR-PE-15

Node: KR-PE-15 | **Vendor:** Cisco | **Severity:** Minor | **Alarm Code:** ALARM_1734

Description

TheCisco PE node, situated at KR Puram Access and denoted by the node ID KR-PE-15, has encountered an optical transport failure. This failure domain, categorized under ALARM_1734 with a minor severity, typically arises due to SFP/XFP degradation, fiber cuts, or Rx/Tx power loss. The PE role, critical for interconnecting multiple sites and facilitating traffic exchange between different autonomous systems, relies heavily on the stability and reliability of its optical transport infrastructure. Underlying this infrastructure are complex protocol and hardware mechanics, including the use of optical fiber cables for data transmission, SFP/XFP modules for signal conversion, and precise power levels for signal integrity. The failure can be attributed to factors such as aging of optical components, physical damage to fiber cables, or misconfigurations leading to power imbalances. Given the critical nature of the PE role in maintaining network connectivity and performance, prompt identification and rectification of the root cause are essential to prevent escalation to more severe failure states and potential network outages.

Pre-requisite Checks

- Dispatch remote hands to KR Puram Access for on-site inspection and intervention if necessary
- Verify redundant link status to ensure network resilience and minimal impact on traffic
- Confirm the availability of spare SFP/XFP modules and fiber cables for potential replacement

Potential Root Causes

Cause	Probability
SFP/XFP Degradation	High
Fiber Cuts	Medium
Rx/Tx Power Loss	Low

Diagnostic Commands & Expected Outputs

```
Command: show interfaces transceiver details, Expected Output:
Detailed information about the SFP/XFP modules, including their type,
wavelength, and power levels, Description: This command isolates
issues related to the optical transceivers, helping identify any power
level discrepancies or module faults that could be contributing to the
optical transport failure.

Command: show ip interface brief, Expected Output: A brief summary of
all IP interfaces, including their status and IP addresses,
Description: This command aids in identifying any interface-related
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issues that could be tied to the optical transport failure, such as an
interface being down due to a fiber cut.
```

Resolution Steps

Step 1: Step: Replace Faulty SFP/XFP Module, Command: replace module with spare, Description: If the diagnostic commands indicate a faulty SFP/XFP module, replacing it with a spare is necessary. This step involves physically removing the faulty module and inserting a new one, ensuring proper seating and securing of the module.

Step 2: Step: Adjust Rx/Tx Power Levels, Command: interface , tx-power , Description: Based on the diagnostic findings, adjusting the Rx/Tx power levels may be necessary to restore optimal signal integrity. This involves using the interface configuration mode to set the appropriate power level for the interface.

Post-Check Verification

```
Command: show interfaces transceiver details, Expected Output: Updated
information reflecting the replacement or adjustment made, with power
levels within the acceptable range, Description: After completing the
resolution steps, this command verifies that the optical transceivers
are operating within the expected power levels, indicating a
successful resolution of the optical transport failure.

Command: show ip interface brief, Expected Output: All relevant
interfaces should be up and operational, with no errors reported,
Description: This command confirms that the network interfaces are
functioning correctly post-resolution, with no signs of the previous
failure.
```

Roll-back Procedures

- Step: Revert to Original SFP/XFP Module, Command: replace module with original, Description: If the replacement module does not resolve the issue or introduces new problems, reverting to the original module may be necessary. This involves physically removing the newly installed module and reinserting the original one.

- Step: Reset Rx/Tx Power Levels to Default, Command: default interface , Description: If adjustments to the power levels do not yield the desired outcome, resetting the interface to its default configuration, including power levels, can help revert any changes made during the resolution attempt.